Anomaly detection for the monitoring of tree dieback from multispectral satellite images

The case of dark beetle infestations

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Outline

Bibliography and references

Monitoring of bark beetle infestations from satellite images
- Spectral indices
- Phenology of vegetation: the dynamics of CR_{SWIR}
- Modeling the expected dynamics of CR_{SWIR}
- Anomaly detection

Bibliography and references

- "Détection des problèmes sanitaires des forêts par imagerie satellite optique: principes et application au cas des scolytes de l'épicéa" by JB Féret, R Dutrieux, K Ose & F de Boissieu in October 4, 2022 at Atelier Theia (https://www.theia-land.fr/foret/2022-sante-foret/) and on the associated publication (https://www.onf.fr/onf/+/cec::les-rendez-vous-techniques-de-lonf-no69-70.html),
- https://waldschutz.wsl.ch/fr/foret/maladies-nuisibles-perturbations/organismes-nuisibles-a-la-sante-des-forets/le-typographe/

Bark beetle infestation

- Infestations are fostered by events such as **storms or droughts**, which knock down or weaken trees.
- The intensive boring by larvae and young beetles beneath the bark **disrupts the flow of sap** in the trees. In addition, the beetles are associated with fungi that **clog the vessels that transport water, leading to the tree dieback.**



Figure: Bark Beetle. Source: Wikipedia



Figure: Architecture of the tunnel system for a typograph moth, a typical dark beetle. Source: Wikipedia[1]

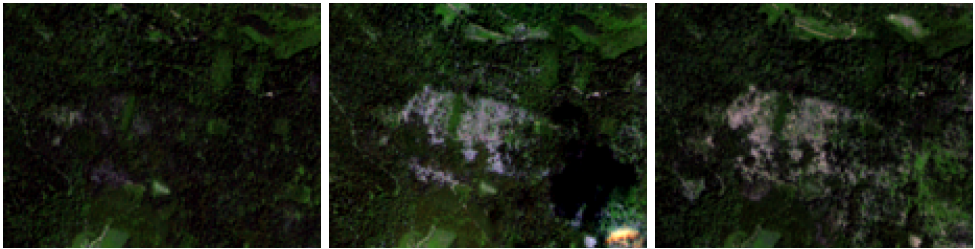


Figure: Effects of bark beetle in the Bavarian Forest. Source: Wikipedia

[1] Meyers Konversations-Lexikon, 4. Aufl. 1888, Bd. 16, S. 352, Ausschnitt aus Tafel Waldverderber I (Käfer); Bd. 16, S. 352 aus Konversations-Lexikon im www[lien cassé], Domaine public, https://commons.wikimedia.org/w/index.php?curid=275883

▸ Remote sensing is highly relevant to **monitor spatial and temporal dynamics** across large forested areas.

▸ Among the wide variety of available sensors, the multispectral instrument of Sentinel-2 offers a very good **balance between spatial resolution, spectral coverage, and revisit frequency**.



(a) 2018 (b) 2019 (c) 2020

Figure: RGB composition of Sentinel-2 images (ESA, Copernicus) - Val d'Ajol, France

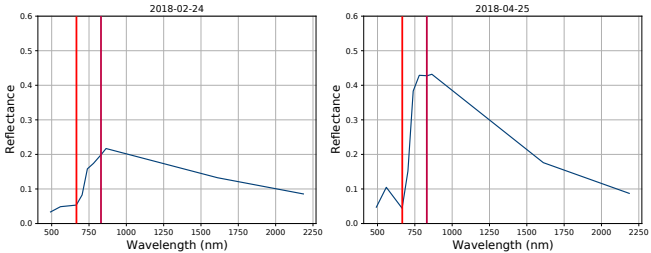
Reflectance provides information about the properties of vegetation

▸ The reflectance estimated from Sentinel-2 images provides information about many properties of the vegetation. In particular, some **spectral indices, i.e. normalized combination of bands, are highly correlated with some leaf traits**.

▸ For instance, the Normalized Difference Vegetation Index (NDVI) is correlated to the chlorophyll content of vegetation and to the structure of plant cover:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{R}}}{\rho_{\text{NIR}} + \rho_{\text{R}}}$$

where ρ_{NIR} and ρ_{R} denote the reflectance in the near infrared (B8 of S2) and in the red (B4 of S2), respectively.



(a) NDVI = 0.55 (b) NDVI = 0.78

Reflectance provides information about the properties of vegetation

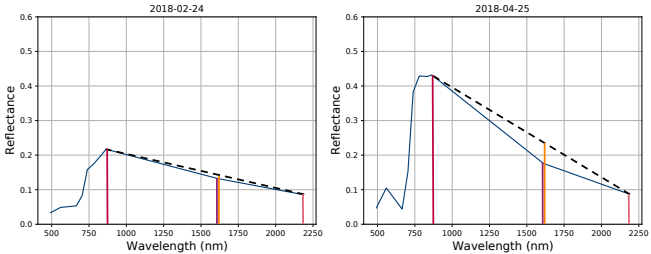
- The spectral index CR_{SWIR} (Continuum Removal in Short-Wave InfraRed) is correlated to the **water content of vegetation**:

$$CR_{SWIR} = \frac{\rho_{SWIR1}}{\rho_{continuum}}$$

where ρ_{SWIR1} denotes the reflectance at 1610 nm (B11 of S2), and

$$\rho_{continuum} = \rho_{SWIR1} + (\lambda_{SWIR1} - \lambda_{NIR2}) \frac{\rho_{SWIR2} - \rho_{NIR2}}{\lambda_{SWIR2} - \lambda_{NIR2}}$$

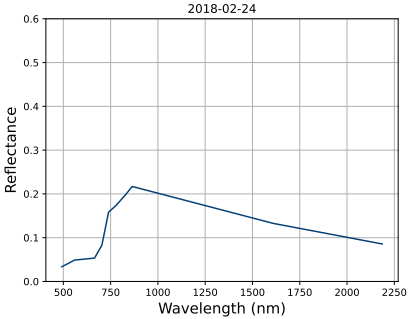
ρ_R denote the reflectance in the near infrared (B8 of S2) and in the red (B4 of S2), respectively.

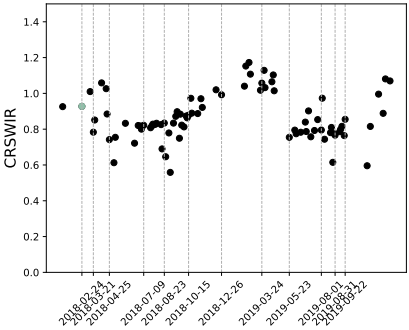


(a) $CR_{SWIR} \approx 1$ (b) $CR_{SWIR} < 1$

Phenology of vegetation: the dynamics of CR_{SWIR} on healthy vegetation

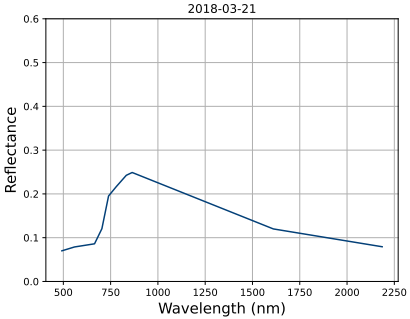


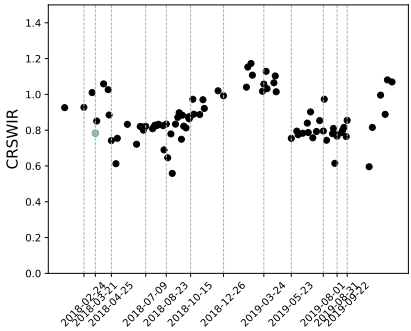




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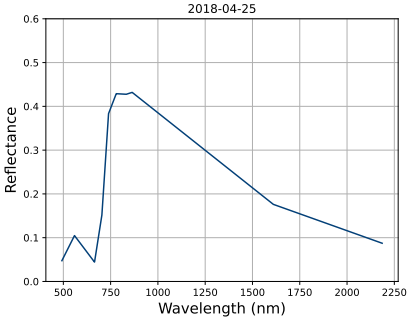


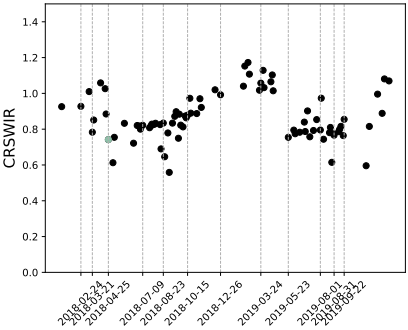




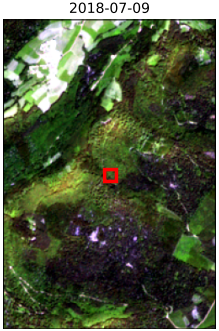
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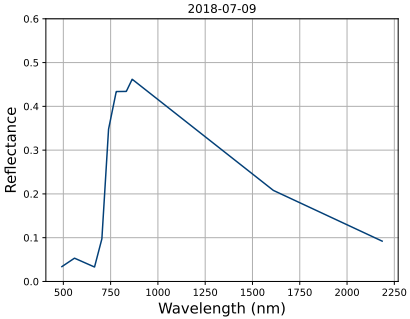


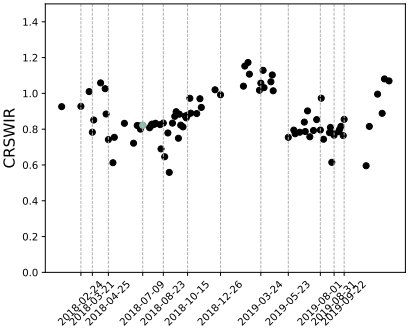




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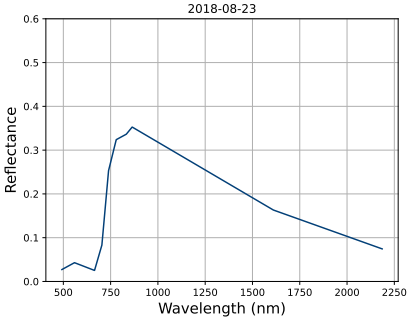


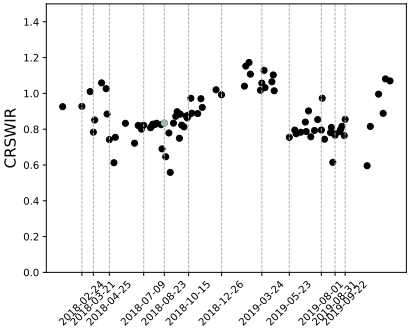




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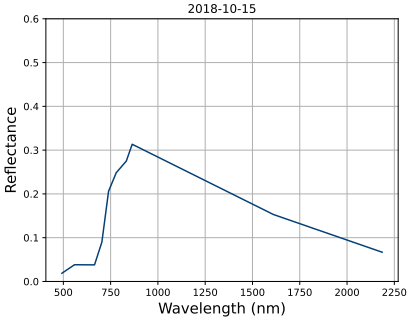


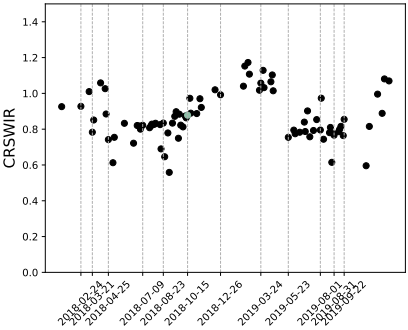




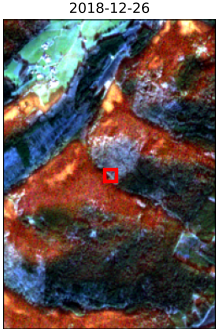
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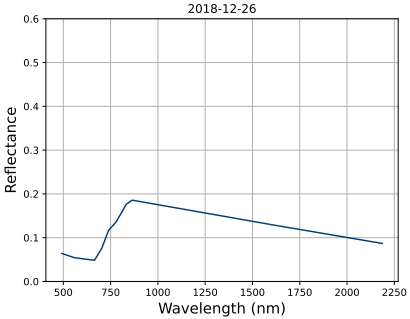


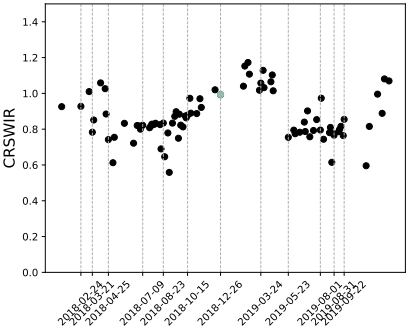




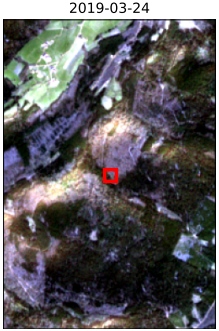
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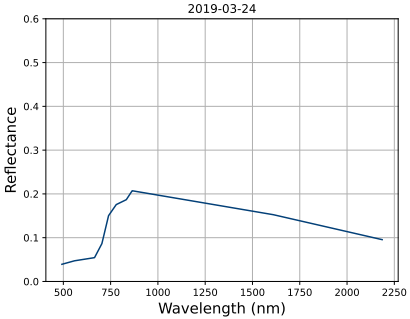


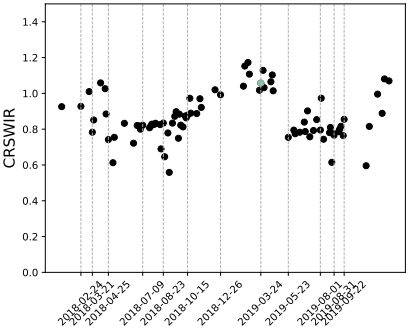




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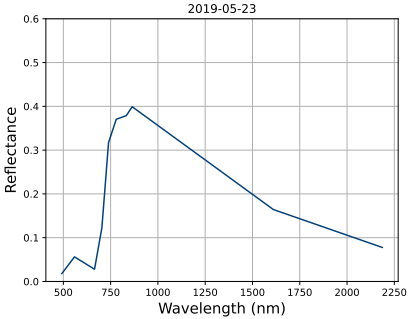


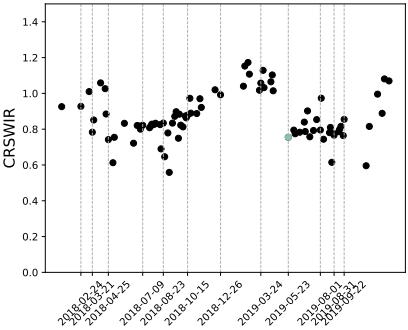




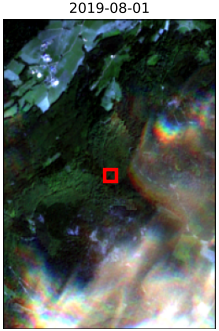
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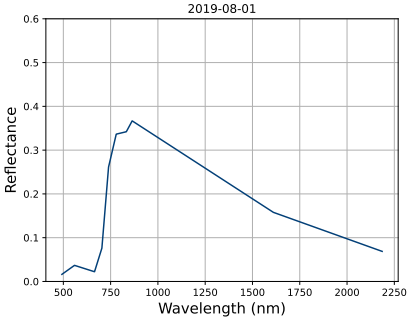


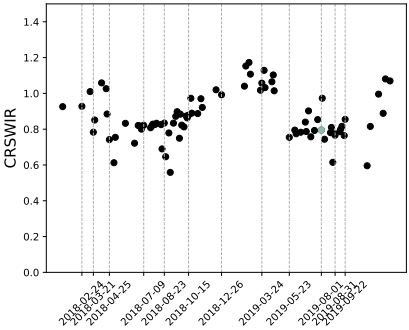




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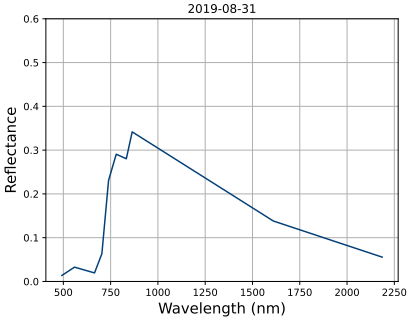


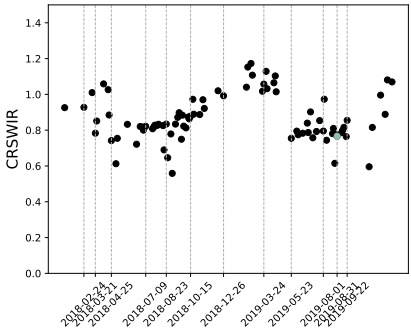




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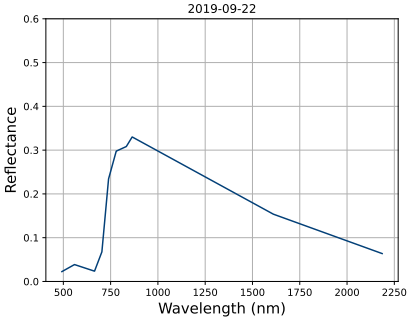


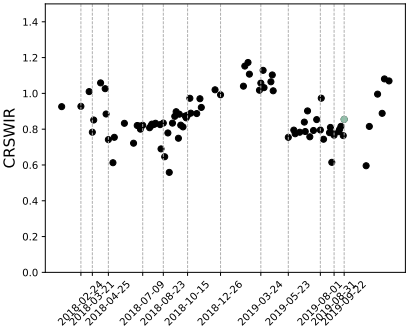




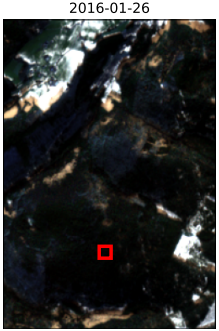
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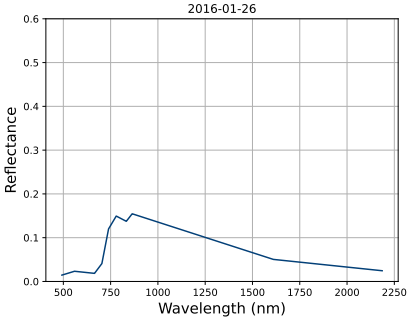


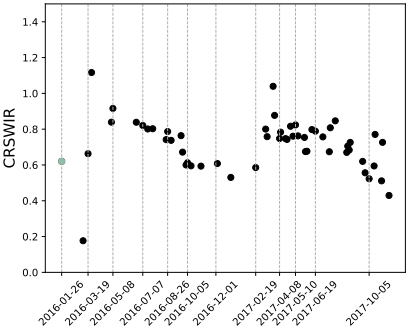




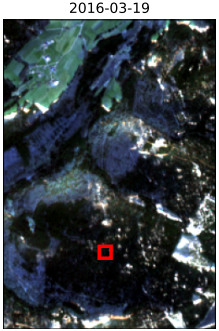
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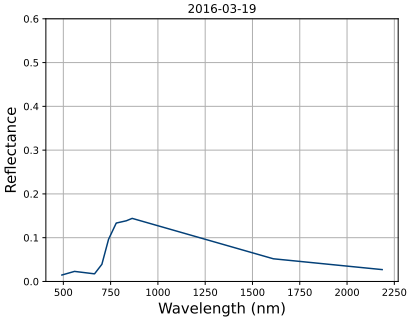


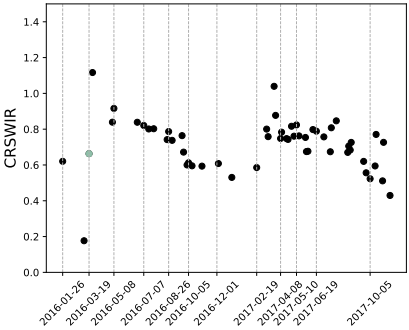




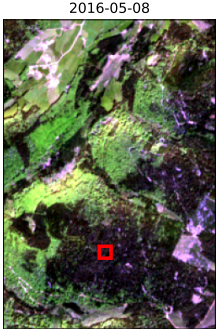
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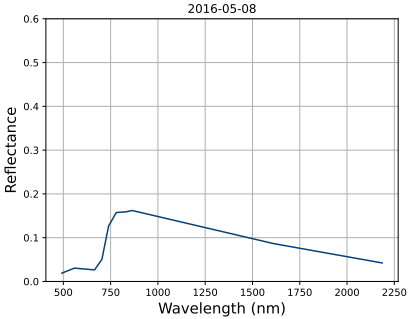


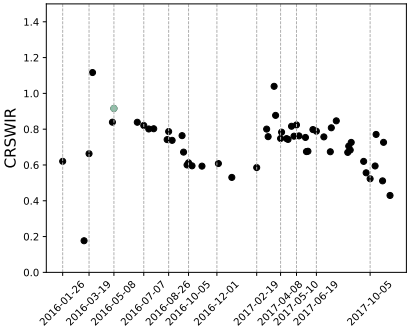




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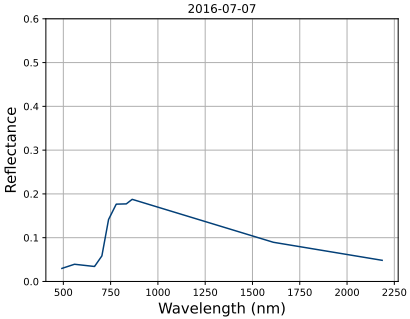


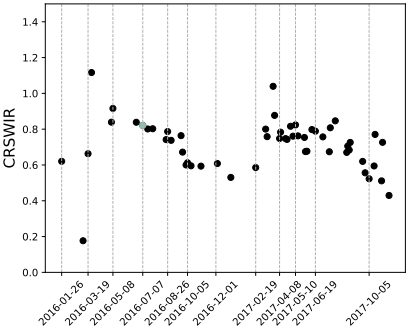




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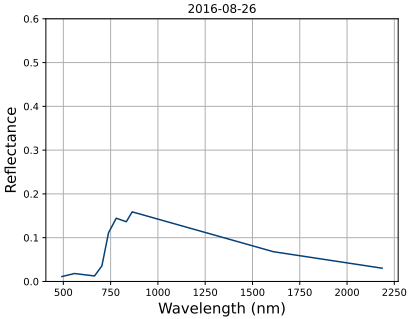


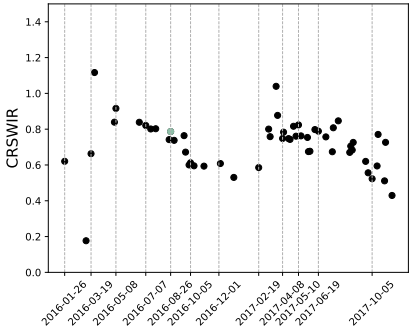




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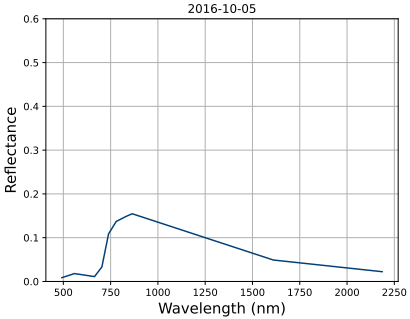


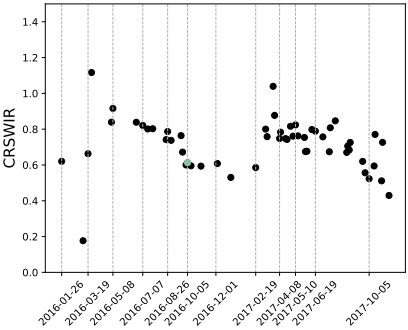




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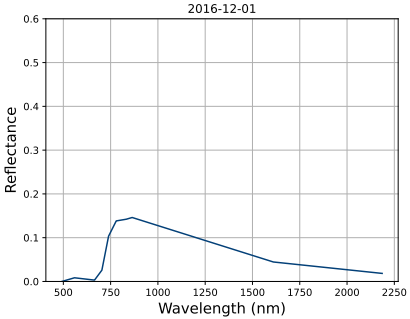


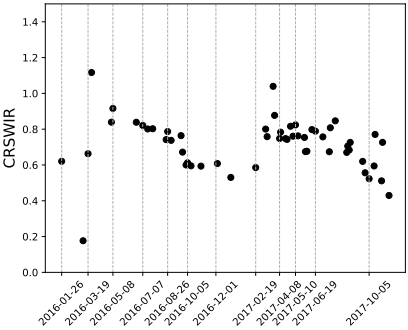




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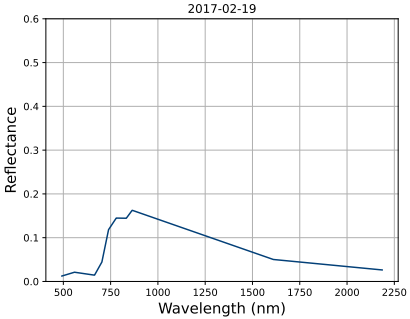


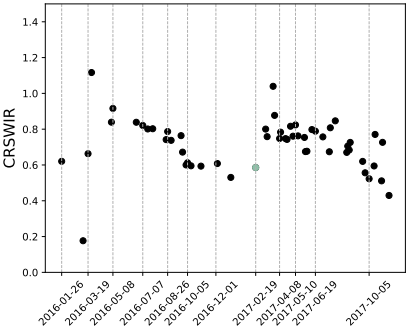




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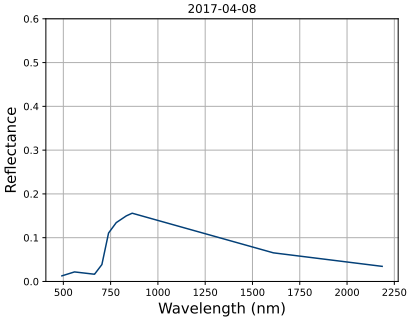


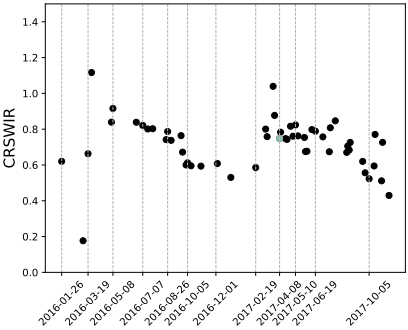




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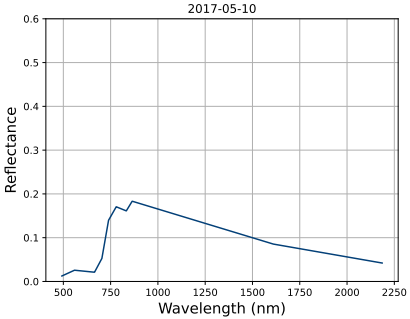


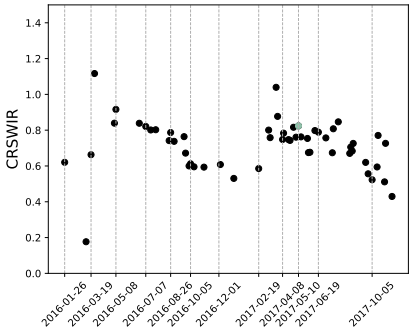




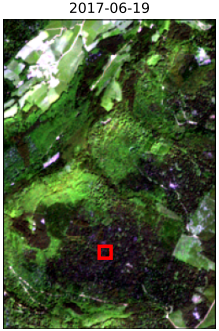
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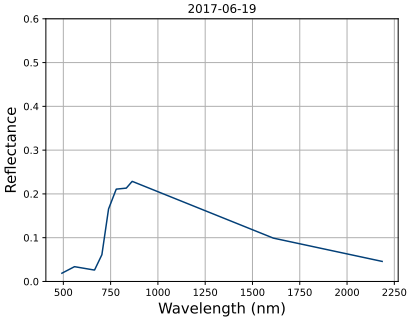


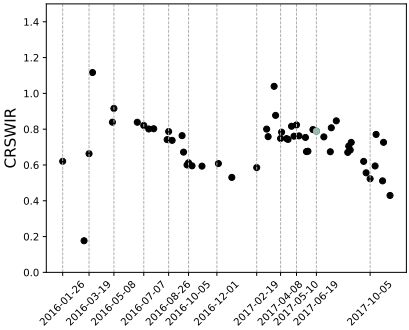




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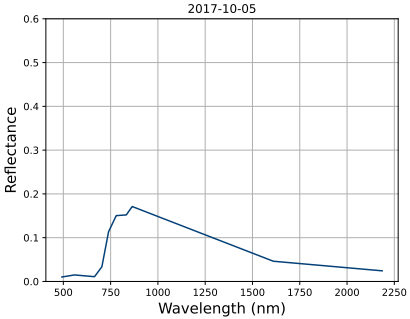


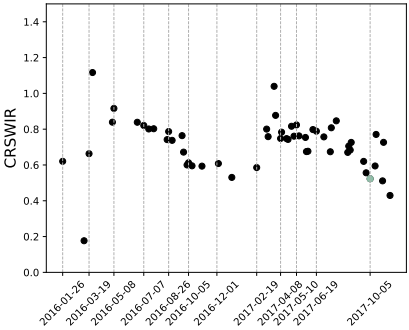




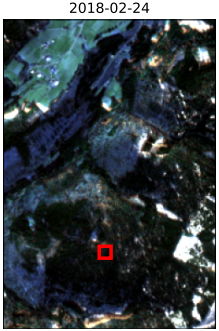
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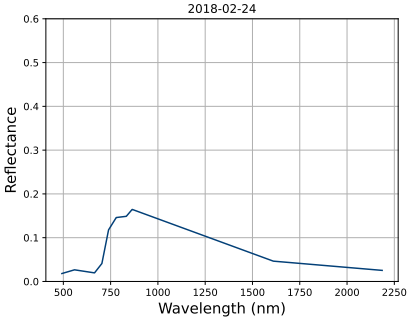


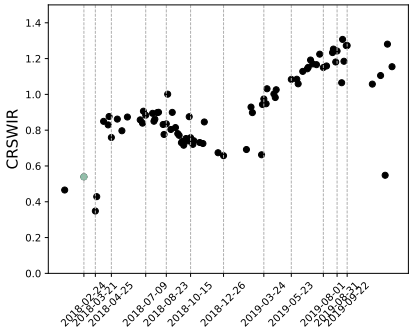




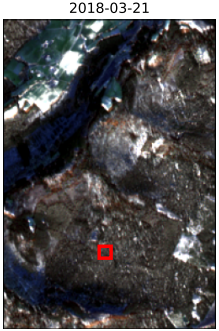
Phenology of vegetation: the dynamics of CR_{SWIR} on infested vegetation

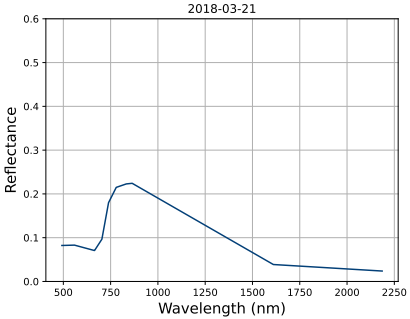


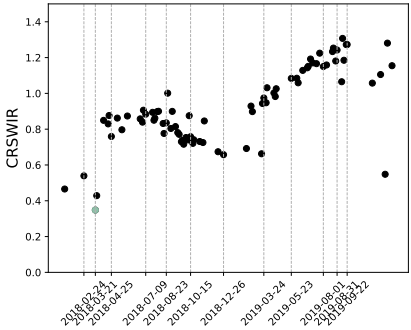




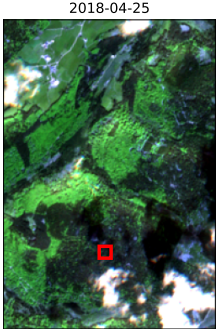
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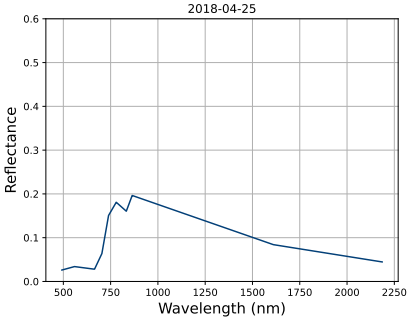


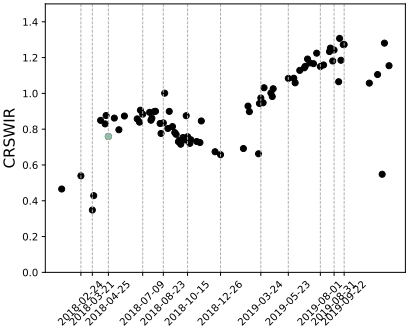




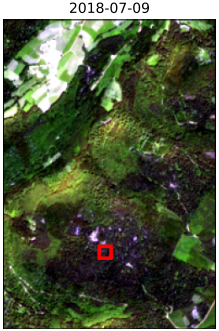
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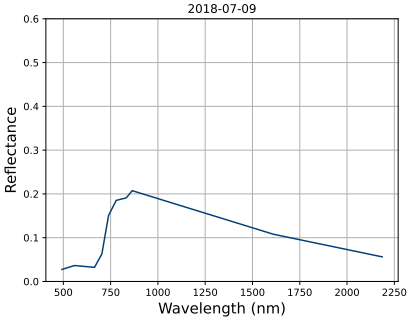


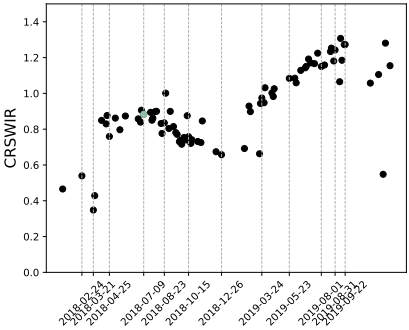




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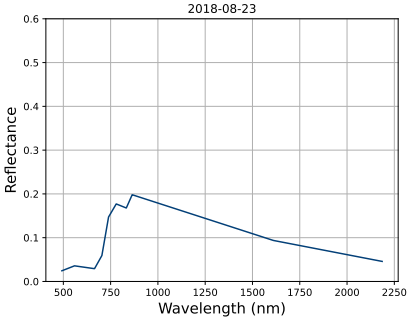


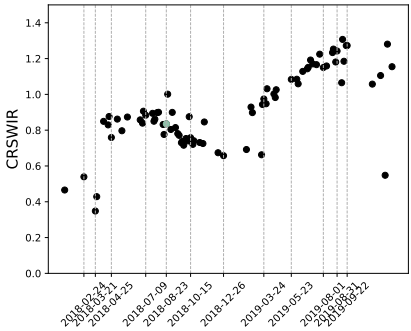




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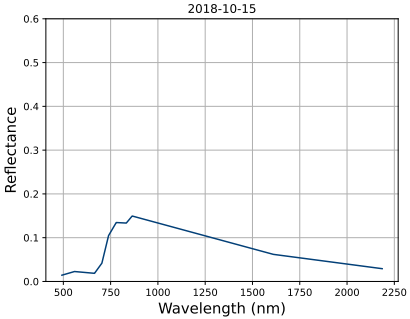


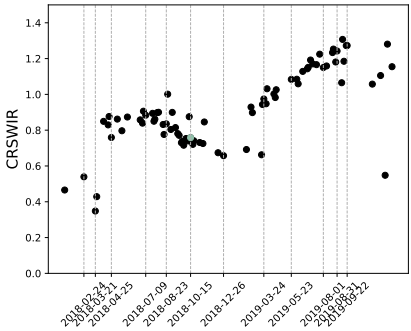




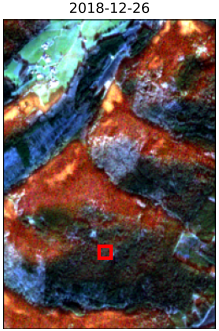
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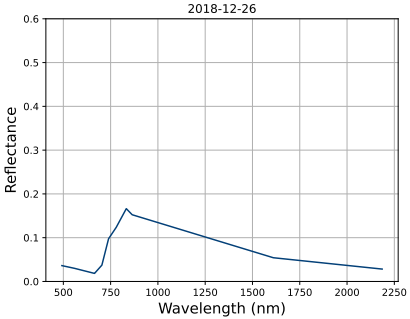


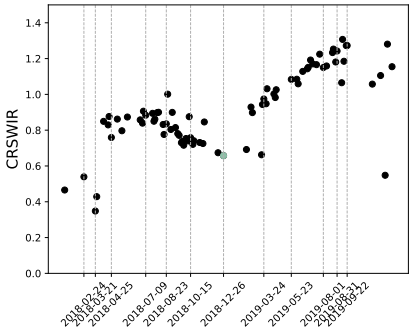




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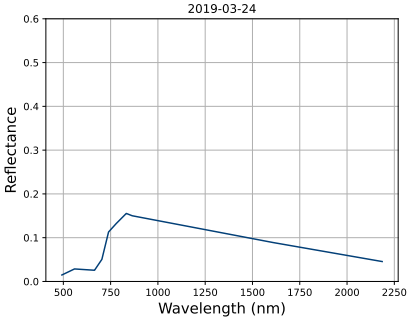


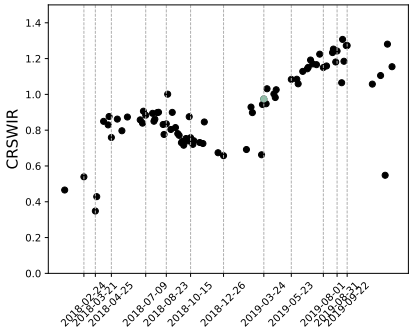




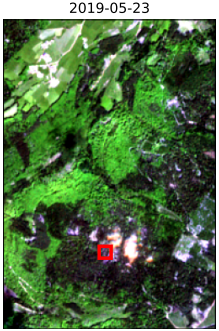
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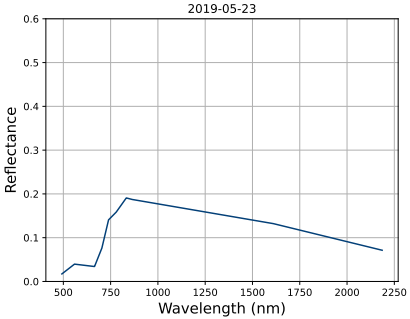


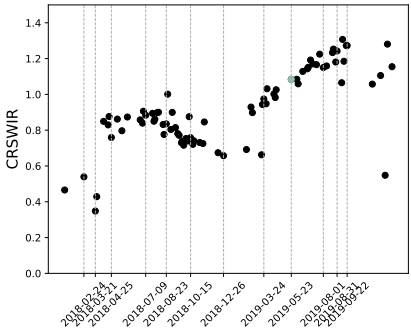




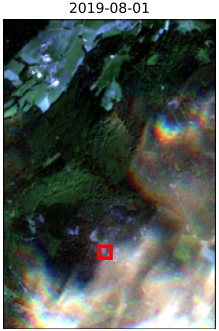
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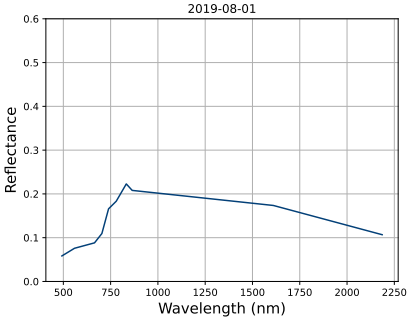


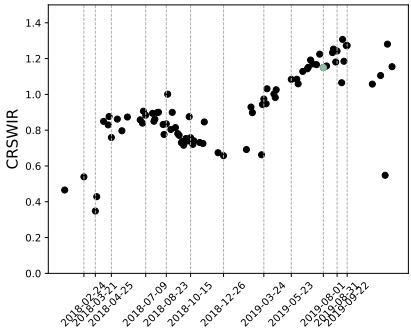




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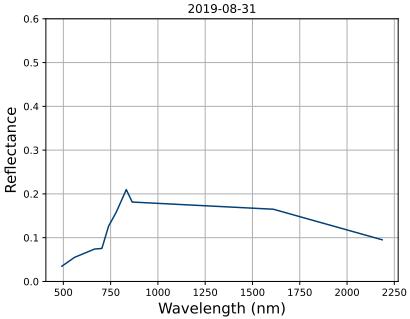


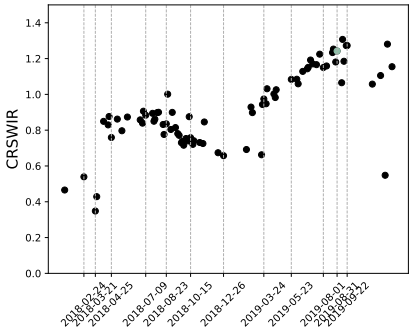




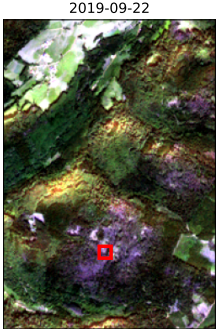
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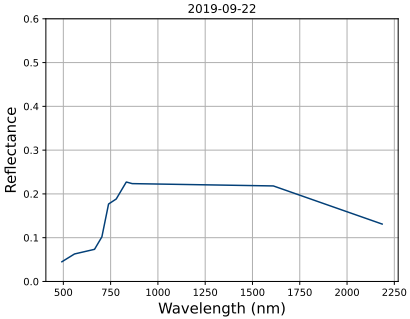


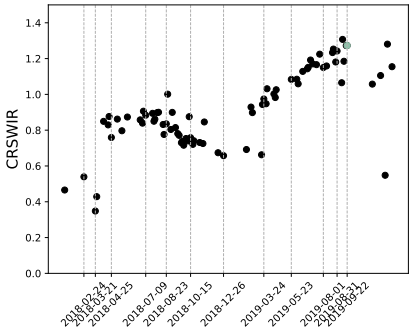




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Modeling CR_{SWIR} with a harmonic model (Féret et al., 2022)

▸ We have seen that the dynamics of CR_{SWIR} for healthy vegetation can be assumed periodic with period $T = 365.25$ days. It is therefore reasonable to model the expected CR_{SWIR} as a partial sum of a Fourier series:

$$\mathbb{E}[\text{CR}_{\text{SWIR}}(\text{day})] = \beta_0 + \sum_{n=1}^{N} \beta_{2n-1} \cos\left(2\pi \frac{n}{T} \text{day}\right) + \beta_{2n} \sin\left(2\pi \frac{n}{T} \text{day}\right).$$

▸ *How can we re-write the expected CR_{SWIR}? How many terms should we keep in the sum?*

Modeling CR_{SWIR} with a harmonic model (Féret et al., 2022)

▸ Let us consider a **2-year CR_{SWIR} time series** as a sample $\{(X_i, Y_i)\}_{(i=1,\ldots,n)}$, where X_i denotes the day of year since a reference date and Y_i denotes CR_{SWIR} at this day.

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- Let's consider the following **linear model**

$$Y_i = \boldsymbol{\beta}^T \boldsymbol{X}_i + \epsilon_i$$

where the errors $\epsilon_1, \dots, \epsilon_n$ are assumed **independent and identically distributed** with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, and $\boldsymbol{X}_i = \begin{bmatrix} 1 & \cos(\frac{2\pi}{T} X_i) & \sin(\frac{2\pi}{T} X_i) & \cos(\frac{4\pi}{T} X_i) & \sin(\frac{4\pi}{T} X_i) \end{bmatrix}^T$ and $\boldsymbol{\beta} = \begin{bmatrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 & \beta_4 \end{bmatrix}^T$.

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Under this model, $\mathbb{E}[Y_i | X_i] = \boldsymbol{\beta}^T \boldsymbol{X}_i$: the harmonic terms capture the **seasonal trajectory** of CR_{SWIR}, and β_0 its annual mean level.

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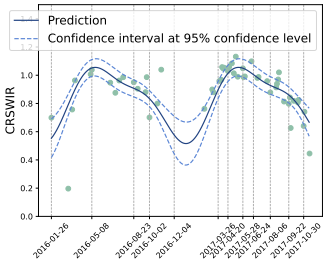
▸ Let us denote by $\hat{\boldsymbol{\beta}}$ the estimator of $\boldsymbol{\beta}$ obtained by ordinary least squares regression. For a day X_0, the estimated expected CR_{SWIR} is $\hat{Y}_0 = \hat{\boldsymbol{\beta}}^T \boldsymbol{X}_0$.

Confidence interval on the expected response

▸ $\hat{Y}_0$ is a random variable and we can show that it follows a Normal distribution with mean equal to $\boldsymbol{\beta}^T \boldsymbol{X}_0$ and variance equal to $\sigma^2 \boldsymbol{X}_0^T (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}_0$, where σ^2 is the variance of the residual error. It follows that a confidence interval at a confidence level $1 - \alpha$ is:

$$\hat{\boldsymbol{\beta}}^T \boldsymbol{X}_0 \pm t_{n-p,1-\alpha/2} \hat{\sigma} \sqrt{\boldsymbol{X}_0^T (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}_0}$$

where $\hat{\sigma}^2$ is the estimated variance of the residual error, n is the size of the sample, p the number of features, and $t_{n-p,1-\alpha/2}$ denotes the quantile of level $1 - \alpha/2$ of a Student distribution with $n - p$ degrees of freedom.



▸ If we had a very large number of samples $\{(X_i, Y_i)\}_{(i=1,\ldots,n)}$, 95% of the confidence intervals constructed using these samples **would actually contain the true model response** $\boldsymbol{\beta X}_0$.

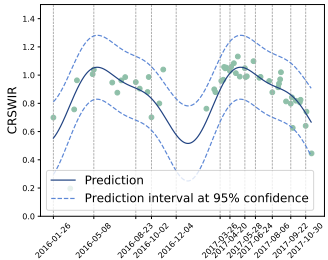
Prediction interval

▸ In order to obtain an interval in which a new Y_0 is expected to fall, we have to take into account the uncertainty on the expected response **and** the uncertainty on the residual error:

$$Y_0 - \hat{Y}_0 = (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \boldsymbol{X}_0 + \epsilon_0.$$

It follows that a prediction interval at a confidence level $1 - \alpha$ is:

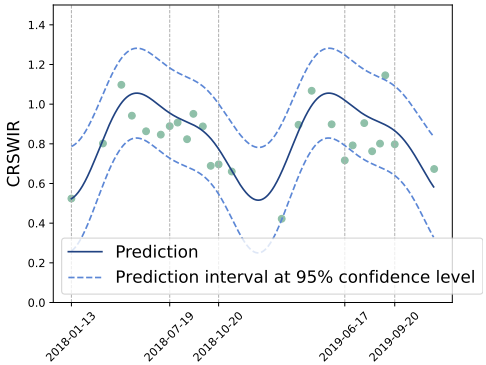
$$\hat{\boldsymbol{\beta}}^T \boldsymbol{X}_0 \pm t_{n-p,1-\alpha/2} \hat{\sigma} \sqrt{1 + \boldsymbol{X}_0^T (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}_0}$$



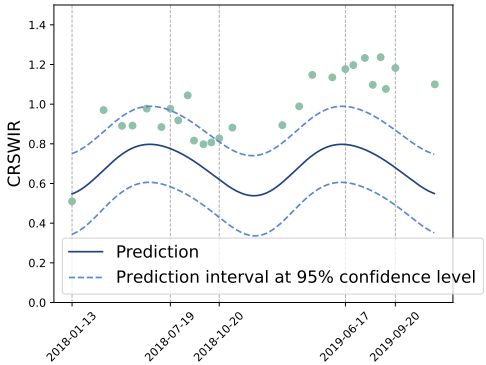
▸ If we observed Y_0 corresponding to a new X_0, and if we had a very large number of samples $\{(X_i, Y_i)\}_{(i=1,\ldots,n)}$, 95% of the prediction intervals constructed using these samples **would actually contain the observation** Y_0.

Tree dieback monitoring as anomaly detection

▸ The prediction intervals hold for healthy vegetation, which makes it a relevant threshold for tree dieback detection!



(a) Healthy vegetation



(b) Infested vegetation

Mapping of tree dieback due to dark beetle infestations

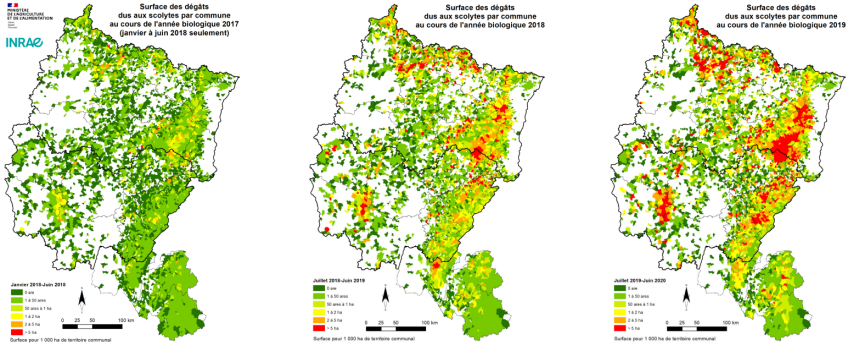


Figure: Dynamics of dark beetle infestations in East of France estimated by (Féret et al., 2022)[1].

[1] JB Féret, R Dutrieux, K Ose & F de Boissieu, Détection des problèmes sanitaires des forêts par imagerie satellite optique: principes et application au cas des scolytes de l'épicéa, October 4 2022, https://www.theia-land.fr/foret/2022-sante-foret/

Let's start coding!